



WORKSHOP

(CARPENTING)

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Introduction

Carpentry involves woodworking tasks like making wooden partitions, roofing, flooring, etc. Common woods used are teak, sal, mango, deodar, and haldu. 'Joinery' refers to making doors, windows, and wooden frames.

Safety Measures

General Safety Rules:

- Pay close attention to what is being done.
- Move carefully when working or climbing.
- Keep the workshop place neat and tidy.
- Sharpen or replace dull tools.
- Disconnect power tools before adjusting them.
- Keep power tools guard in place.
- Avoid interrupting another person who is using a power tool.

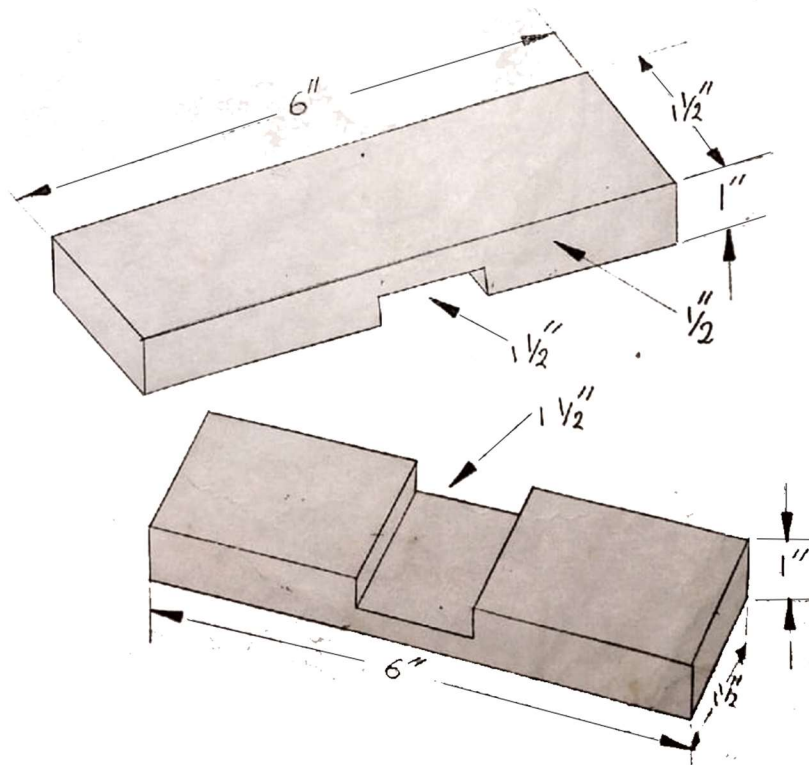
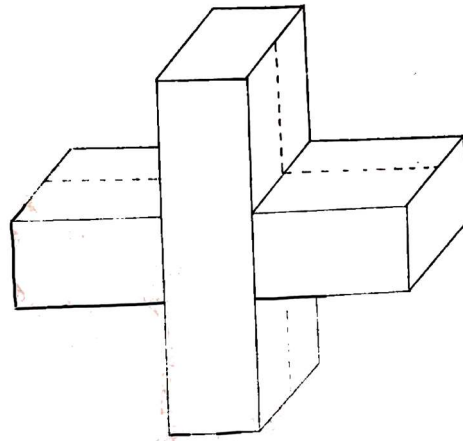
Personal Safety:

- Avoid injury by using safety devices like gloves, aprons, and safety shoes or boots.
- Ensure clothing is tight-fitting and hair is secured.

Experiment: Half-Lap Joint

Aim: To make a half-lap joint.

Material Size: Length = 12", Breadth = 1.5", Thickness = 1".



Tools Required:

- Rip saw
- Tenon saw
- Firmer chisel
- Mortise chisel
- Try square
- Marking gauge
- Rasp file
- Mallet hammer
- Ball-pane hammer
- Flat file
- Sandpaper

Procedure for Making a Half-Lap Joint

1. Setup and Marking:
 - Measure the width of one wood piece and use the combination square to mark this distance on both pieces. The marking gauge ensures consistent line depth, while the pencil highlights the lines for cutting.
2. Cutting the Joints:
 - Secure the wood pieces with clamps to prevent movement. Adjust the circular saw to cut halfway through the thickness. Chisels can also be used for precision cutting and refining edges. Use the try square to verify the accuracy of cuts and ensure the surface is flat and precise.
3. Cleaning the Joints:
 - Remove remaining material using the chisel and mallet for precision. Smooth out the joint surface with sandpaper for a clean fit.
4. Assembling the Joint:
 - Align the pieces using a ruler to ensure a tight fit, and secure with clamps to hold the joint in position.
5. Finishing:
 - Smooth the assembled joint using sandpaper or a file and confirm the fit with the try square to achieve a polished, uniform appearance.